Bala

SKU:K014











Description

Bala is a self-balancing robot application. This product is a self-balancing robot composed of the M5 FIRE and a dual-channel DC motor base. The name "BALA" is derived from the abbreviation of "Balance". It comes pre-installed with a self-balancing car application, which uses a closed-loop algorithm to maintain vertical balance during operation. It uses accelerometer and gyroscope attitude data to correct its orientation and position.

Features

- 2-channel DC driver module
- I2C communication: 0x56
- Compatible with LEGO
- POGO Pin
- Supports TF card expansion

Includes

- 1 x Bala
- 1 x Motor driver
- 2 x N20 motors (with built-in encoders)

- 1 x Type-C USB

Specifications

Specification	Parameters
ESP32	240MHz dual-core, 600 DMIPS, 520KB SRAM, Wi-Fi
Flash	16MB Flash + 4MB PSRAM
Input	5V @ 500mA
Interfaces	Type-C x 1, GROVE(I2C+I/O+UART), Pogo Pin x 1
LCD	2-inch, 320x240 color TFT LCD, ILI9341
Speaker	1W-0928
Microphone	MEMS analog BSE3729 microphone
LED	SK6812 3535 RGB LED x 10
MEMS	BMM150+(MPU6886/SH200Q)
Battery	Built-in 550mAh @ 3.7V
Operating Temperature	32°F to 104°F (0°C to 40°C)
Main Control Size	54 x 54 x 21 mm
C.A.S.E	Plastic (PC)
Net Weight	130g
Gross Weight	247g
Product Size	90 x 54 x 61mm
Package Size	185 x 108 x 81mm

Learn

BMM150 Magnetic Field Interference

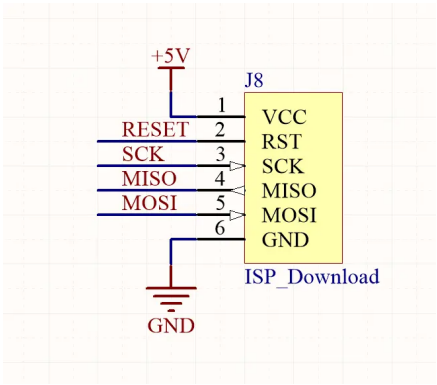
Products with magnets may interfere with the BMM150 magnetic field sensor, causing abnormal readings. When used with an M5 master control device containing a magnet, the magnet needs to be removed, and at the same time, the BMM150 sensor should be kept away from strong magnetic fields.

Sensor Calibration

Note: Calibration is required before first use! Hold the rightmost C key to power on, release the key after hearing a "beep" sound, and the sensor will enter calibration mode. Keep the host level and stationary, and the sensor calibration will be completed in 3 seconds. After calibration, it will automatically enter balance mode. If BALA cannot maintain balance during use, try calibrating the sensor to resolve the issue.

PinMap

Mega328 ISP Download Interface Pin Definition



Softwares

Arduino

- Bala Balancing Example.

EasyLoader

Easyloader	Download Link	Notes
Bala Test Firmware MPU6050 Version Easyloader	download	/
Bala Test Firmware MPU6886 Version Easyloader	download	/

Video

BALA Demonstration

M5BALA .mp4

BALA Demonstration - Phone Control

Iphone Controlled M5Bala .mp4

BALA Demonstration - Line Following

M5BALA.mp4

BALA Demonstration - Line Following with Phone

Self-tracing Car.mp4